April 21, 1959

PROJECT 7000

FILE MEMO

SUBJECT: The Dependence of STRETCH Internal Computing Speed on Various Configuration Parameters

I. Introduction

There have recently been a number of requests for more information concerning the performance of STRETCH. Most of the requests have been for performance figures for STRETCH systems with other than 6 memory units and for an estimate of the effect on computing speed of Input-Output devices running at the same time.

The graphs presented here are based on Timing Simulator runs made on the IBM 704. Some of the points are from runs made especially for this report, others are based on various studies made last year.

It should be emphasized that the results presented in these graphs are smoothed and must be considered as approximate representations of how STRETCH will behave on typical examples. The actual percentages can vary considerably depending on the particular problem configuration being done.

II. Speed vs. Number of Look-Ahead Levels

The Look-ahead serves as an internal buffer during the preparation and execution of instructions. Its main job is to allow STRETCH to use the overlapping asynchronous organization built into the computer design. It is a key factor in permitting a match of the high speed arithmetic circuits to the relatively slow memory units by multiplexing. It also has the important feature of serving as a "virtual memory" to avoid refetching repeated quantities from the main memory.

Graph I shows how the speed of STRETCH varies for one typical problem assuming different numbers of look-ahead levels in the design. "No levels" implies that the arithmetic unit is tied directly to the instruction preparation unit.

III. Speed vs. Number of Memory Units

IV. The Effect on Internal Computing Speed of Simultaneous I/O Activity

V. The Effect of the High Speed Disk

The effect of the High Speed Disk on STRETCH Speed: (Approximate reduction in speed of internal computation for systems with different numbers of memory units caused by disk running at the same time.)

This indicates that the user of a small system is penalized twice: once by a reduction of the top speed of his system and again by a larger I/O penalty when it is run concurrently with the computation.

In all fairness, however, the above reductions in performance with I/O running are still far preferable than stopping CPU activity completely as the 704 does, particularly when one contrasts the information flow rates involved.

H. G. Kolsky

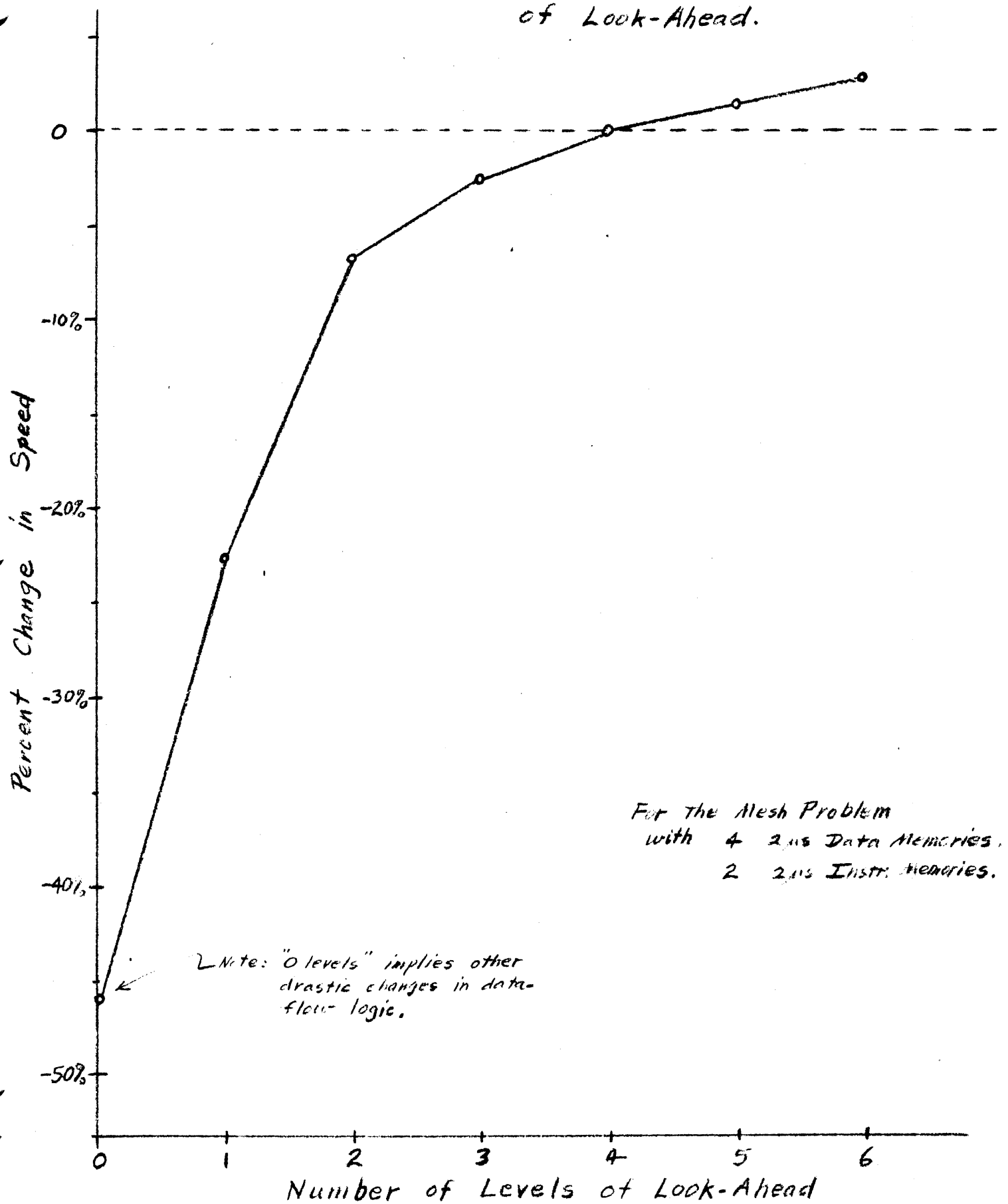
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HGK:jcj
Enclosures: 5 Graphs

Graph I

(Mesh Prob.)

STRETCH: Speed of Internal Computation vs. Number of Levels of Look-Ahead.



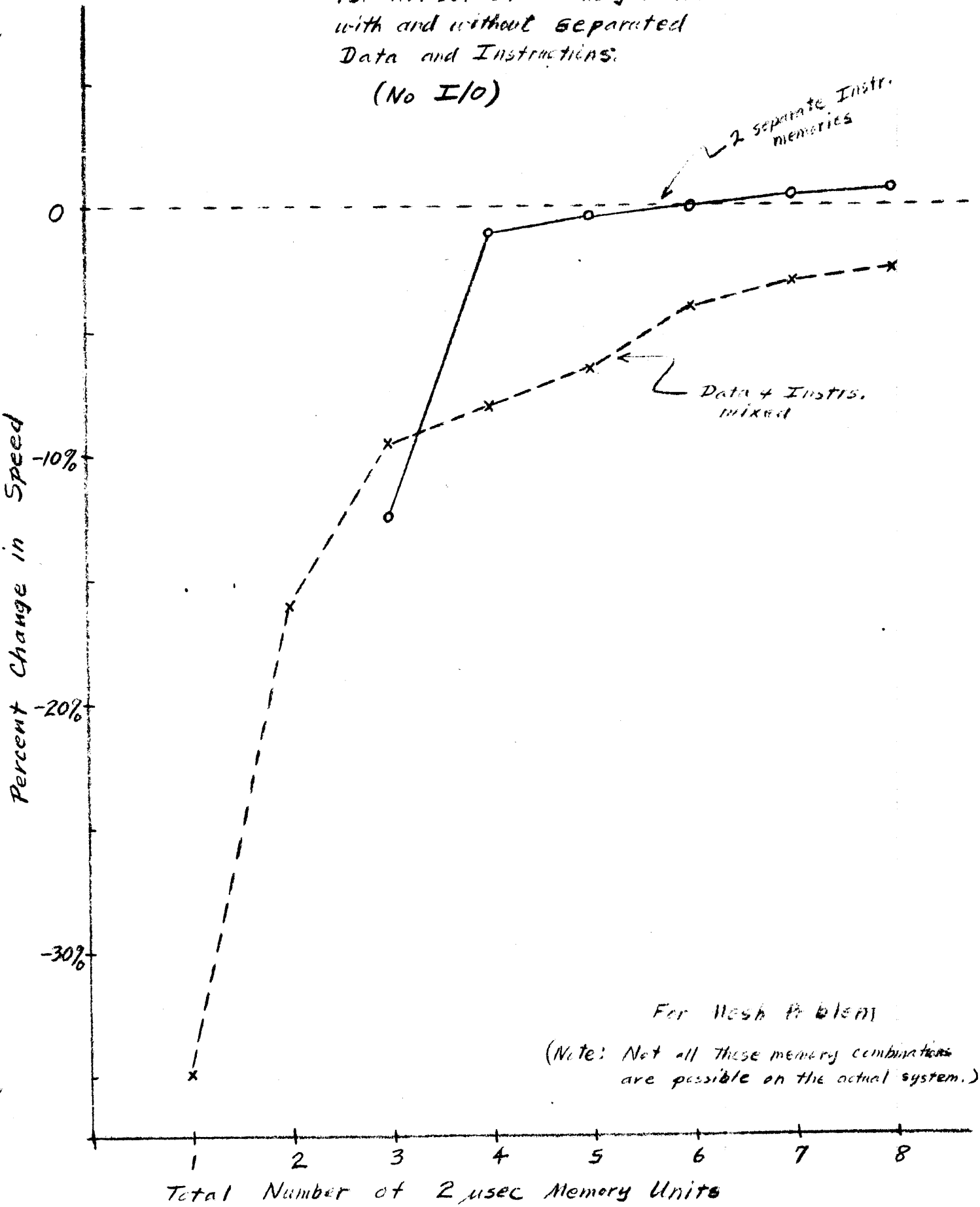
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Graph II

(Mesh Prob.)

STRETCH: Speed of Internal Computation vs. Number of Memory Units with and without Separated Data and Instructions.

(No I/O)



For Mesh Problem

(Note: Not all these memory combinations are possible on the actual system.)

HGN Apr 9, '59